Network Cables & Crimping Assignment

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Q1. Cat6 Ethernet Cable Diagram & T568B Wire Color Standard

Below is the pinout and wire arrangement diagram for an RJ-45 connector using the industry-standard **T568B wiring standard** on a Cat6 Ethernet cable:

🔌 RJ-45 CONNECTOR T568B PINOUT DIAGRAM (CAT6)

Pin 1	White / Orange (TX+)	Transmit Data (+)
Pin 2	Solid Orange (TX-)	Transmit Data (-)
Pin 3	White / Green (RX+)	Receive Data (+)
Pin 4	Solid Blue	Unused / Telephony
Pin 5	White / Blue	Unused / Telephony
Pin 6	Solid Green (RX-)	Receive Data (-)
Pin 7	White / Brown	Power over Ethernet (PoE) / Reserve
Pin 8	Solid Brown	Power over Ethernet (PoE) / Reserve

Q2. Three Network Cable Types & Real-World Applications

Comparison of cable types, their specs, and real-world implementation scenarios:

Cable Type	Max Bandwidth / Distance	Real-World Application Device	Why it is chosen
Cat5e	1 Gbps up to 100 meters	**Zomato POS Terminal / Billing Counter**	Sufficient speed for lightweight transaction data; highly cost-effective for basic LAN connectivity.
Cat6	10 Gbps up to 55m / 1 Gbps at 100m	**Wi-Fi 6 Router / Gaming Consoles**	Higher shielding and internal spline reduce cross-talk; ideal for low-latency gaming & fast LAN streaming.
Fiber Optic Cable	100+ Gbps over tens of kilometers	**IPL Streaming Server / ISP Backbone**	Transmits data as light pulses; unaffected by electromagnetic interference and offers massive bandwidth for 4K video live streaming.

Q3. Step-by-Step RJ-45 Crimping Process & Common Mistakes

Step-by-Step Crimping Guide:

1. **Strip Outer Jacket:** Use a wire stripper to remove about 1 to 1.5 inches of the outer PVC jacket without nicking inner wires.
2. **Untwist & Straighten Wires:** Separate the 4 twisted pairs, cut off the plastic spline separator (for Cat6), and flatten wires neatly.

3. **Arrange in T568B Order:** Align wires left-to-right: *White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown*.
4. **Trim to Size:** Trim the aligned wires cleanly using crimping tool cutters so they extend roughly 0.5 inches (1.2 cm).
5. **Insert into RJ-45 Connector:** Push wires firmly into the connector with the locking clip facing down until copper tips touch the front edge.
6. **Crimp Connector:** Insert into the crimp tool's 8P slot and squeeze handles firmly to push gold pins into wire conductors. Test with a LAN cable tester.

> ⚠️ **Two Common Mistakes to Avoid:**
> 1. **Stripping Outer Jacket Too Far Back:** Leaving outer cable jacket outside the connector clip causes wire strain and easy breakage.
> 2. **Wrong Wire Order or Incomplete Insertion:** Swapping wire positions or failing to push wires to the front edge results in open circuits or failed pin connections.

Q4. Gaming Zone & IPL Screening Setup Scenario

Recommended Cable & Connector Setup:

- **Cable Type: Cat6a (or Cat6) Shielded Twisted Pair (STP) Ethernet Cable**
- **Connector Type: Shielded Metal-Clad RJ-45 Connectors**
- **Main Feed / Server Link: Single-Mode Fiber Optic Cable** for main high-speed stream router feed.

Explanation & Justification:

- **Low Latency for Gaming Consoles:** Cat6/Cat6a provides up to 10 Gbps speeds with low latency, essential for lag-free multiplayer gaming on PS5/Xbox consoles.
- **Electromagnetic Interference (EMI) Protection:** In a gaming zone with multiple 4K TV screens, heavy power cabling, and consoles, **STP shielding** prevents signal interference.
- **Uninterrupted IPL 4K Live Stream:** Fiber optic internet feed to the primary router ensures zero buffering during live 4K match streaming.

Q5. Real-World Network Cable & Connector Observation

📸 Observation: Blue Cat6 Patch Cable with Clear RJ-45 Connector

Connected behind a Wi-Fi Router / PC Ethernet Port

- **Cable Type: Cat6 UTP (Unshielded Twisted Pair) Ethernet Patch Cable** with a molded blue boot.
- **Connector Type: Clear Plastic 8P8C RJ-45 Modular Plug** with gold-plated contact pins.
- **Devices Connected:** Connects the LAN port of a **Wi-Fi Router** directly to the **Ethernet Network Interface Card (NIC)** of a Desktop PC/Laptop or Smart TV for high-speed internet access.